

oricdemo2026

A HIRES-mode demoscene production for the Oric Atmos (6502A @ 1 MHz, bare metal, no ROM calls), built with the Oscar64 cross-compiler. A nod to the machine's own 1985 "Welcome to Oric Atmos" tape demo — same spirit (a splash, a logo, an animated scene, a handful of effect showcases, a credits roll), reworked as a real multi-section production with real music, real converted photographs, and a few original techniques of its own.

Ships as **two independent, fully-playable distributions** built from the same source (`src/main.c` + `src/section_*.c`):

- **LOCI** (`build/oricdemo.tap` + 8 asset files) — loads via a LOCI mass-storage device. The `.tap` itself is only the code — music and every picture load from disk at runtime, so **all 9 files must sit together in one folder** on the LOCI's SD card/USB storage (see Installation below). Requires a real LOCI device or an emulator that supports one; plain tape/cassette playback or LOCI-less emulation will load the program but leave every picture/music load silently failing.
- **Floppy disk** (`build/oricdemo_floppy.dsk`) — a single, fully self-contained bootable Microdisc disk image with everything baked in. No LOCI, no DOS/SEDORIC, nothing else needed at all — just boot it.

Both play the full demo end-to-end, with real AY-3-8912 music (Arkos Tracker) and 12 sections, cycling forever once started. **If you just want to watch/listen without any real or emulated LOCI hardware, the floppy image is the simpler option.**

The demo

In running order:

1. **idi8b splash** — a per-cell dissolve-in/out brand wordmark.
2. **Oric logo + raster bars** — the ORIC ATMOS 48K wordmark with two rotating, colour-cycling highlight bars sweeping behind/in front of it.
3. **Bird scene** — a parallax sky/creek backdrop with clouds and an animated bird flying a sine-wave path (a direct nod to the original 1985 demo's own animated bird).
4. **HIRES shapes showcase** — ellipse, star (outline + flood-fill), pattern-fill, and flood-fill primitives, one at a time.
5. **Polygon workout** — a continuously rotating, pulsing wireframe star.
6. **3D function surface** — a rotating wireframe height-field mesh, projected through a real 3D camera pipeline.
7. **Sprite showcase** — a satellite sprite drifting over a starfield.
8. **Scroll showcase** — a byte-aligned hardware-charset text scroller over a hand-illustrated Oric Atmos desk scene.
9. **Wave showcase** — a vintage magazine photo, sine-distorted in place.

10. **Macaw showcase** — a full-colour scarlet macaw photograph with a scrolling caption.
11. **Raster IRQ showcase** — three full-screen colour bars, each driven directly from a 50Hz hardware interrupt, sweeping over three procedurally-drawn stars (filled, outline, and hatched).
12. **Credits** — a longer-form scroll over a converted sunset photograph, crediting the tools, music, and source material used throughout.

Two music tracks (Arkos Tracker `.aky` modules) alternate automatically whenever the current one finishes a full playthrough.

See `docs/architecture.md` for the full technical writeup (memory maps, per-section techniques, subsystem design) and `docs/README.md` for the library reference manual behind `include/`.

Installation

Floppy disk (simplest — no LOCI needed)

Boot `build/oricdemo_floppy.dsk` under Microdisc emulation (or write it to a real floppy for real Microdisc hardware) — that’s it, everything’s baked into the one image.

LOCI

The LOCI distribution is **9 files that must all live together in one folder** on the LOCI’s SD card/USB storage — the program loads music and every picture from that same folder at runtime (relative to wherever the `.tap` itself was launched from, same convention as this project’s sibling `OricScreenEditorLOCI`):

Filename	Description
<code>oricdemo.tap</code>	Main program
<code>steppingout.aky</code>	Music track 1
<code>boulesetbits.aky</code>	Music track 2
<code>oriclogo.bin</code>	Oric Atmos logo picture
<code>starfield.bin</code>	Starfield picture
<code>oricatmos.bin</code>	Oric Atmos desk-scene picture
<code>oricmag.bin</code>	Vintage magazine-photo picture
<code>macaw.bin</code>	Scarlet macaw photo
<code>sunset.bin</code>	Credits-screen sunset photo

`make zip` builds all of these into one release ZIP, already laid out at `idi8b/oricdemo2026/` inside the archive (the same `idi8b/<ApplicationName>/` distribution convention this project’s own `.env.example` documents for `USBPATH`) — plus the floppy `.dsk` and a PDF of this README at the archive’s top level (unrelated to the LOCI folder requirement).

To install:

1. Unzip the release ZIP directly onto the LOCI device's SD card/USB storage root — it already contains the right `idi8b/oricdemo2026/` folder with all 9 files above together in it, no manual reorganizing needed.
2. Go to the LOCI user interface (press the LOCI's own button), select tape images (**T**), browse (**SPACE**) to that folder, and select `oricdemo.tap` with **SPACE**.
3. Boot into the program with **ESC**. If Auto Load is enabled on the LOCI, it starts automatically; otherwise type `CLOAD""` and press **RETURN**.

For details on operating the LOCI device itself, see the LOCI User Manual.

A LOCI device (real or emulated) is required — plain tape/cassette playback, or an emulator with no LOCI support, will load and run the program, but every music/picture load will silently fail (graceful degradation, not a crash): you'll get a mostly-blank, silent demo. See “Building from source” below for which emulators actually emulate LOCI.

Building from source

Requires Oscar64 and, for testing/emulation, Oricutron and/or Phosphoric. Python tooling (`tools/oric_pictconv.py`, `tools/oric_ttfconv.py`, `tools/oric_floppybuilder.py`) needs Pillow: `pip install -r tools/requirements.txt`.

```
make          # -> build/oricdemo.tap (LOCI target)
make disk     # -> build/oricdemo_floppy.dsk (floppy target)
make run-disk # launch the floppy target in Oricutron -- fully self-contained,
              # including Oricutron's own debugger/monitor (F2, breakpoints)
make run-phos # launch the LOCI target in Phosphoric, WITH LOCI emulation
              # (--loci-flash assets) -- the full experience, real AY audio too
make test     # automated regression suite (Phosphoric, headless)
make zip      # release ZIP: LOCI target + all 8 assets (under idi8b/oricdemo2026/) +
              # floppy .dsk + README.pdf (both at the ZIP's top level) -- see Installation
make usb      # copy the LOCI target + all 8 assets to a USB stick/LOCI SD card
```

(There's no plain Oricutron target for the LOCI/tape build — Oricutron has no LOCI emulation, so it could only ever show a silently-degraded demo. `run-disk` already gives the same Oricutron debugger access with a fully working demo instead.)

`OSCAR64_HOME` defaults to `~/oscar64`, `ORICUTRON_HOME` to `~/oricutron` if unset. `USBPATH/PHOSDIR` are read from `.env` (copy `.env.example` to `.env` and fill in values). See `CLAUDE.md` for the complete build-target reference (every `make` target, both runtimes, test fixtures) and `docs/README.md` for the `include/` library API.

Credits

- **Code:** Xander Mol (xahmol), built with Claude Code.
- **Music:** “Stepping out” (2019) — Mr.Lou, Dewfall Productions. “Boules Et Bits” — Tom & Jerry. Both Arkos Tracker modules.
- **Bird sprite:** adapted from mihai-dragan’s oric_BAS project (MIT License).
- **Oric Atmos logo:** Oric International’s own wordmark, via Wikimedia Commons.
- **Scarlet macaw photo:** Kandukuru Nagarjun (Jurong Bird Park), CC BY 2.0.
- **Credits-screen sunset photo:** Artem Beliaikin, via Wikimedia Commons, CC0.
- **A nod to** “Welcome to Oric Atmos” (Oric International, 1985) and to idi8b.

Every external asset’s exact source/license is also credited in a header comment at its own point of use in `src/`.